

## Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
  - (a) a
  - (b) Chemistry
  - (c) 0
  - (d) 1
2. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
  - (a) Condition codes set by every instruction
  - (b) Variable-length encoding of instructions
  - (c) Instructions requiring widely varying numbers of cycles to execute
  - (d) Several different classes (sets) of registers
3. In Python 3, which of the following function removes all leading and trailing whitespace in string?
  - (a) `replace(old, new [, max])`
  - (b) `strip([chars])`
  - (c) `swapcase()`
  - (d) `title()`
4. In Python 3, what is the following function returns item from the list with max value?
  - (a) `cmp(list)`
  - (b) `len(list)`
  - (c) `max(list)`
  - (d) `min(list)`
5. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
  - (a) The goal of the attack
  - (b) The number of computers being attacked
  - (c) The number of computers launching the attack
  - (d) The time period in which the attack occurs

## True / False (10 marks)

*Answer True or False*

6. True or False: The minimum value of the list  $l = [1,2,3,4]$  in Python 3 is 1.
7. True or False:  $O(N^{1/4})$  functions grow slower than  $O(N^{1/N})$  functions in the context of algorithmic complexity.
8. True or False: Bitmap graphics allow for the accurate representation of all line segments as straight lines without distortion.
9. True or False: The ratio of nonterminal nodes to total nodes in a complete K-ary tree of depth N is approximately  $(K-1)/K$ .
10. True or False: A<sub>3</sub>-way, set-associative cache allows each main memory word to be stored in any of 3 designated cache locations.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a python function to count the number of rectangles in a circle of radius r.
12. Write a python function to find even numbers from a mixed list.

*End of Paper*